RC Car Hardware Assembly Manual

RC Car Assembly Manual Series No.1

This manual is the first one of our RC car assembly manual series. In this manual, we will explain the hardware assembly of the RC car. Please read and follow the instructions carefully. There may be changes in parts due to future discontinuation issues.

RC Car Assembly Manual Series

0. Battery Safety Rules Manual
1. RC Car Hardware Assembly Manual
2. VESC Setup Manual
3. Jetson Xavier NX Setup Manual
4. Keyboard Control with ROS2
5. RC Car Calibration Manual

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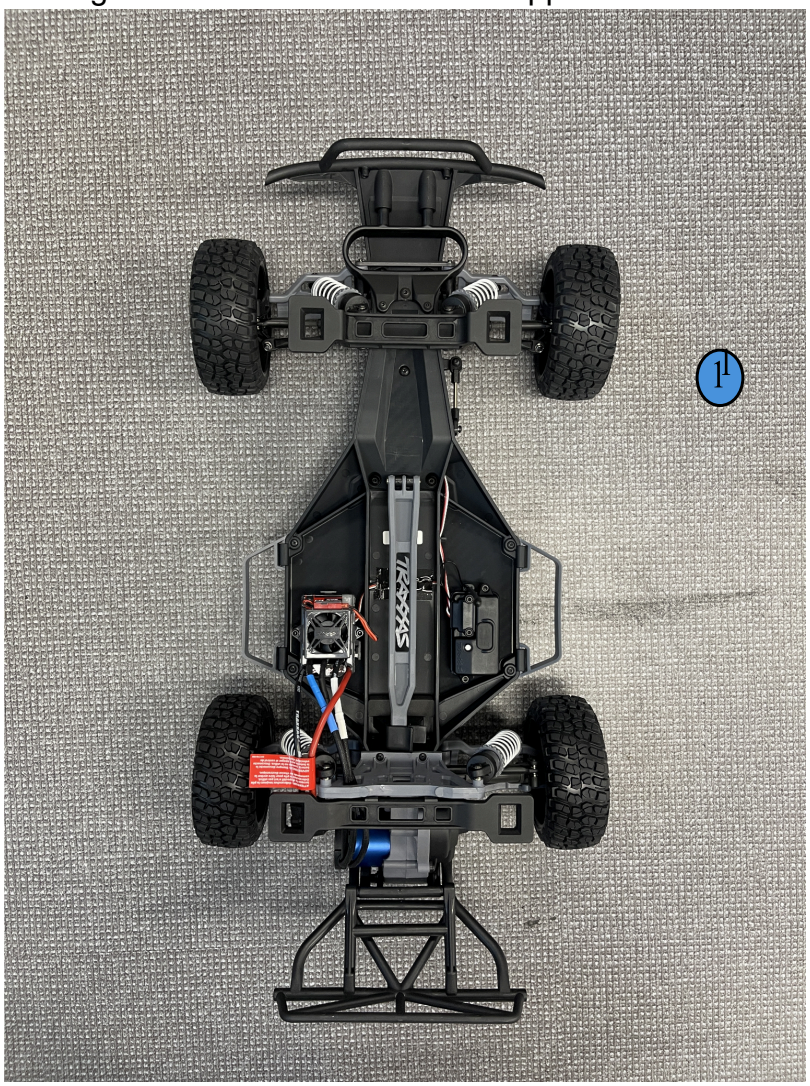
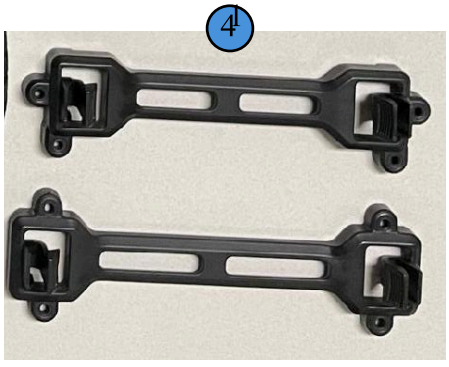
    5-3. Actual product names and purchase links

0. Ingredients

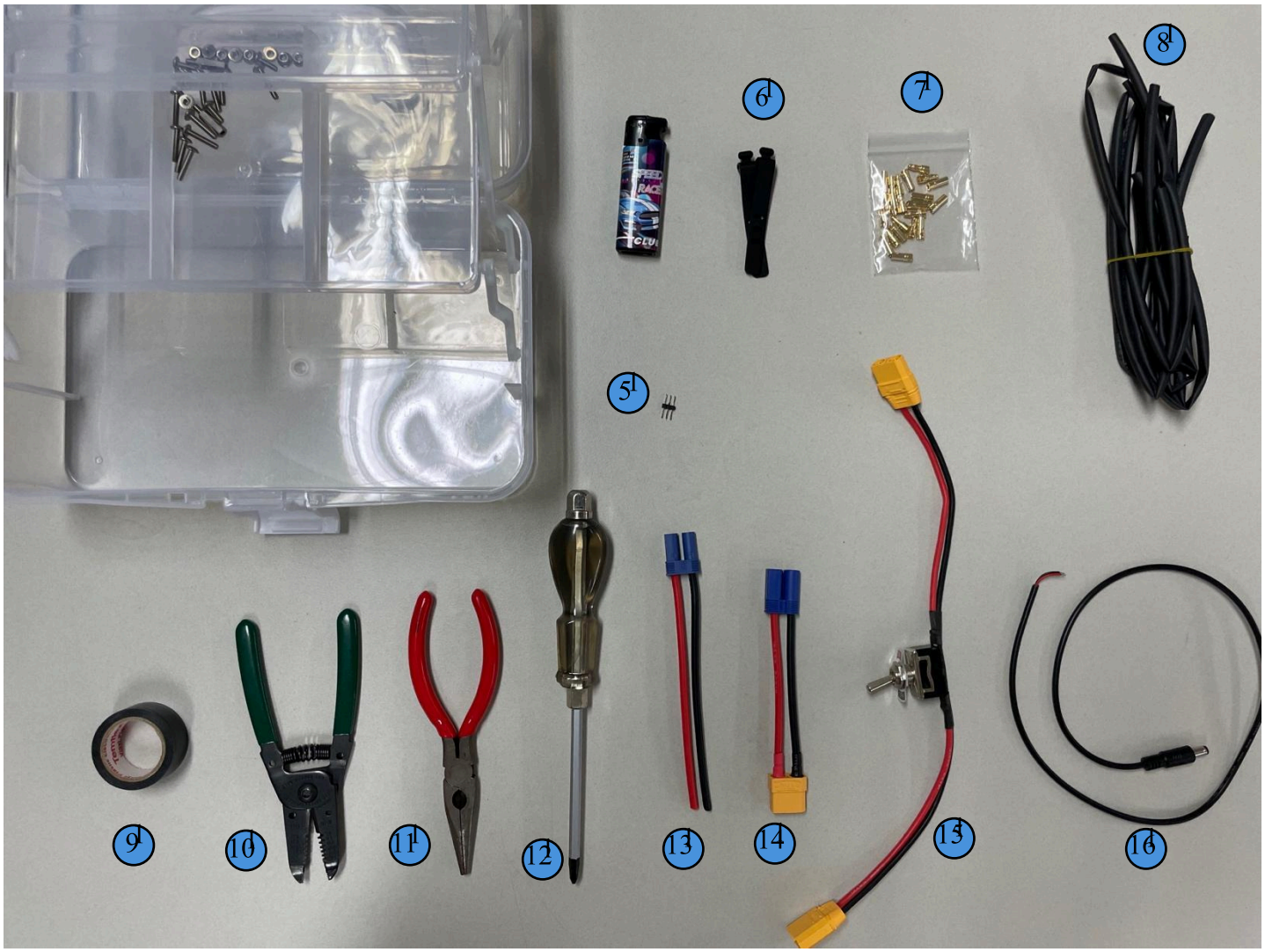
0-1. Ingradients: RC car chasis and upper deck

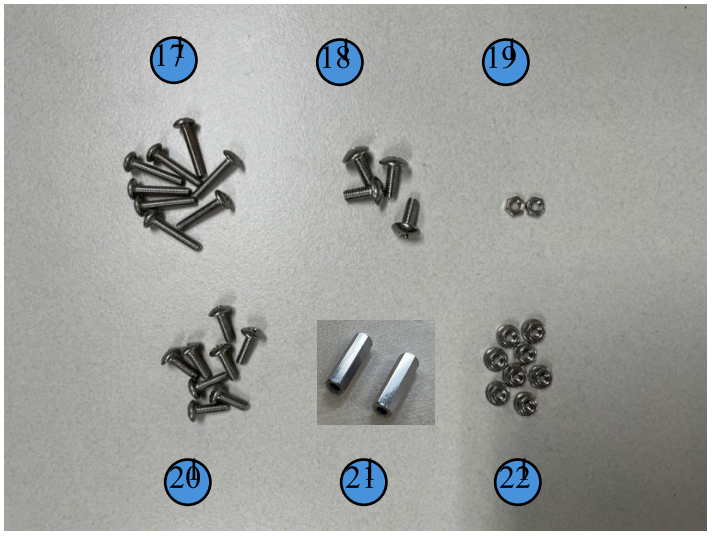


1	Traxxas Slash 1/10	3	Jetson fixer
2	Upper deck	4	Chasis-deck connecting component



0-2. Ingradients: Tool box





5	3 pin connector	14	EC5 to XT90 cable

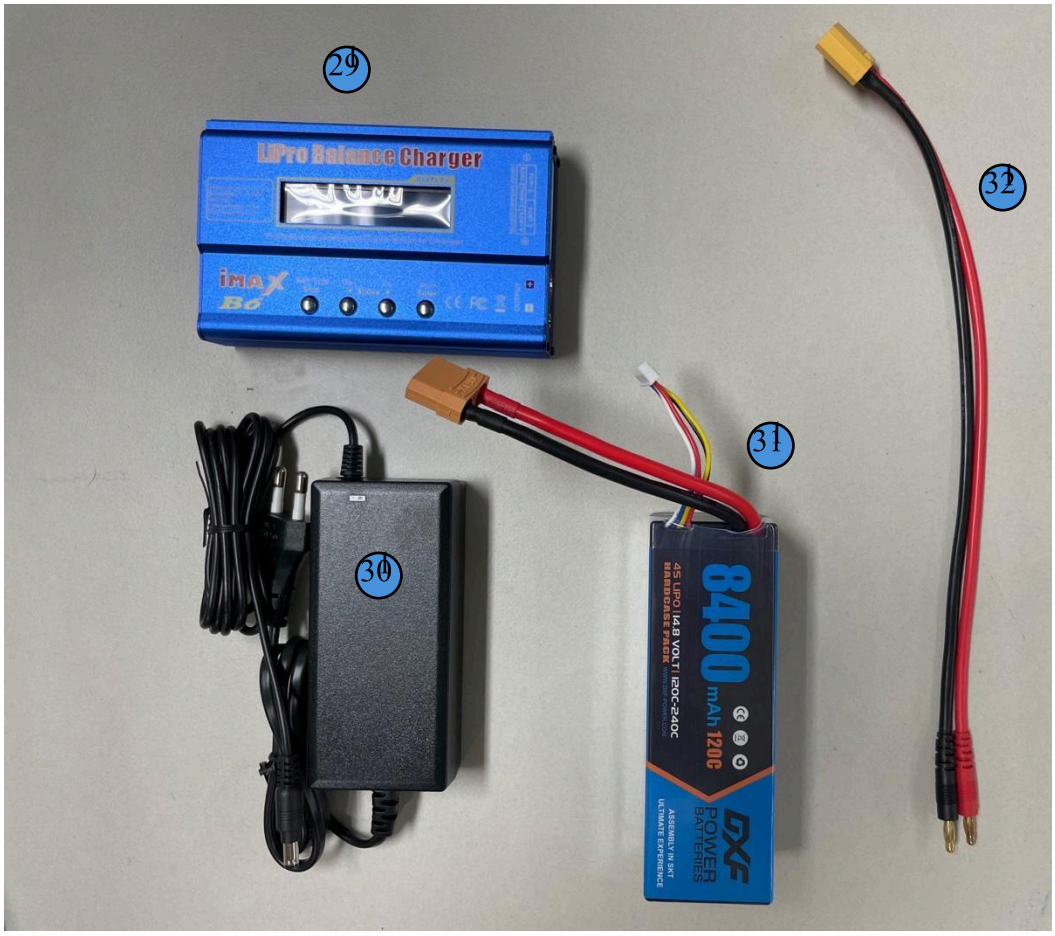
6	Silicon cable tie	15	XT90 switch cable
7	3.5mm gold jack	16	Power longing cable
8	Heat sink tubes (2/3/4/5 mm) + lighter	17	M3 16mm screw (x8)
9	Insulation tape	18	M4 10mm screw (x4)
10	Wire stripper	19	M3 flat nut (x2)
11	Long needle nose plier	20	M3 10mm screw (x10)
12	Screwdriver	21	M3 15mm spacer (x2)
13	EC5 longing cable	22	M3 closing nut (x8)

0-3. Ingredients: main components

23	HDMI cable	26	Power distribution board
24	Micro 5 pin cable	27	FSESC 6.7 pro
25	NVIDIA Jetson Xavier NX board	~~28~~	~~Velineon 3500 brushless motor~~

0-4. Ingradients: LiPo battery charge modules

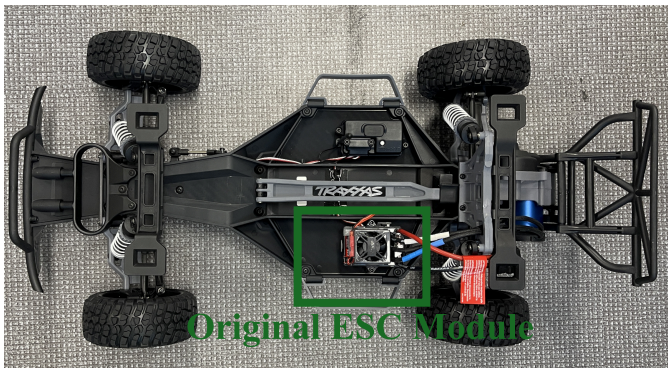
(Refer to the supplementary material for the method of charging the LiPo battery.)



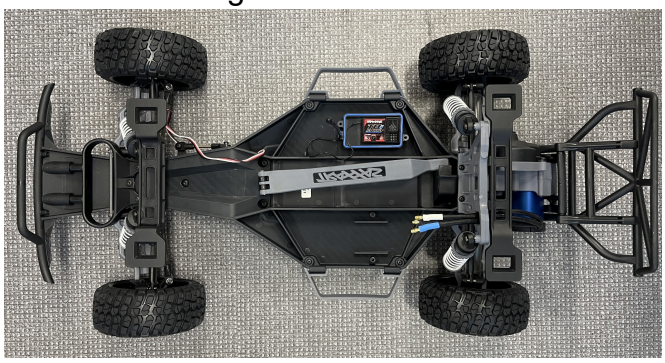
29	Imax B6 balance charger	31	4S LiPo battery
30	Battery charger power cable	32	Battery charger connector(banana plug)

1. Remove unnecessary components

1-0. Naïve car

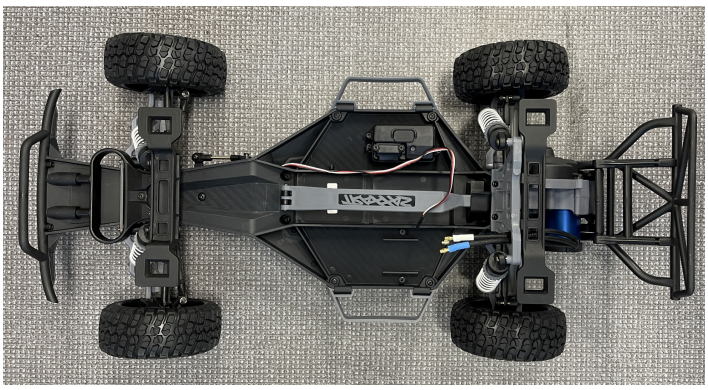


1-1.Abandon original ESC module



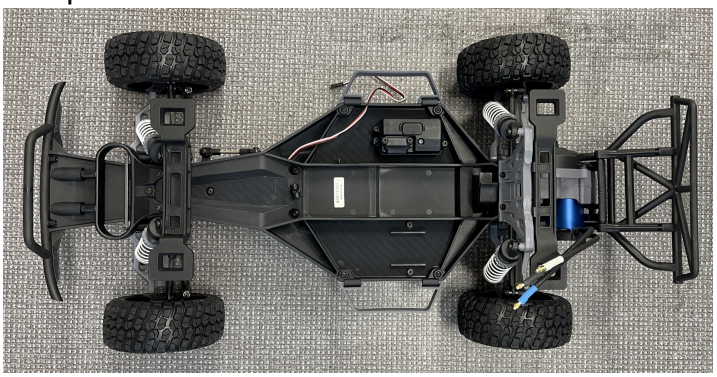
1-2. Remove the receiver antenna stick and neatly organize the internal wires. (Please keep them well organized, as they will be used again next semester)

1-3. Reattach and secure the receiver cover. .



1-4. Remove the central battery holder. (It can be removed more easily by pulling it with long-nose pliers while turning the hex driver.) Abandon original ESC module.

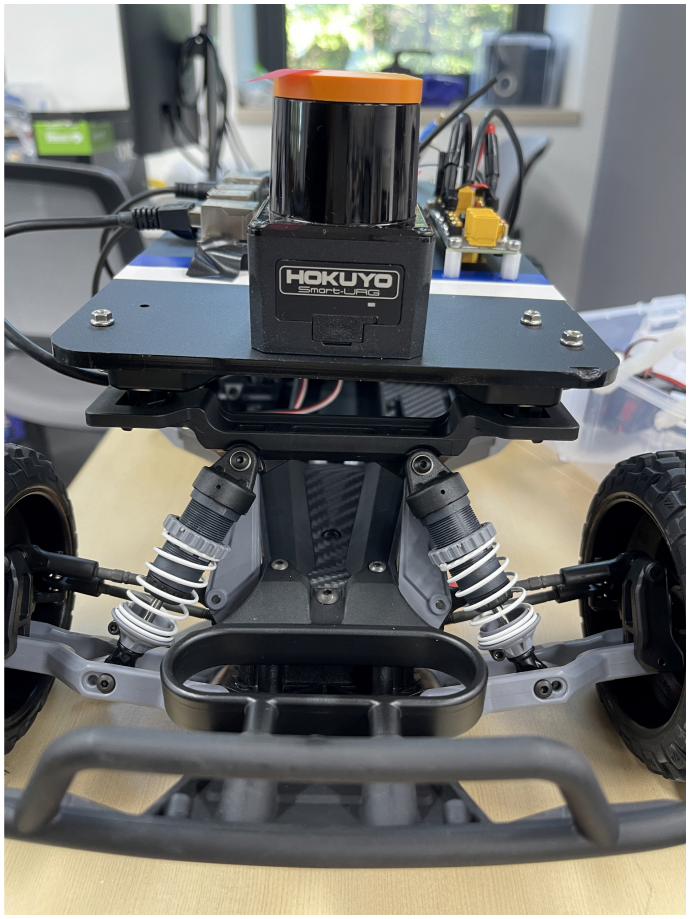
Completed State:



2. Spring Length Adjustment

2-1. There are spring-damper suspensions at the front and rear. The spring length can be adjusted by turning the adjustment ring. Since this adjustment affects driving performance and vehicle leveling, adjust it according to your preference. An example is shown below.

Front:



Rear:

3. Upper deck assembly

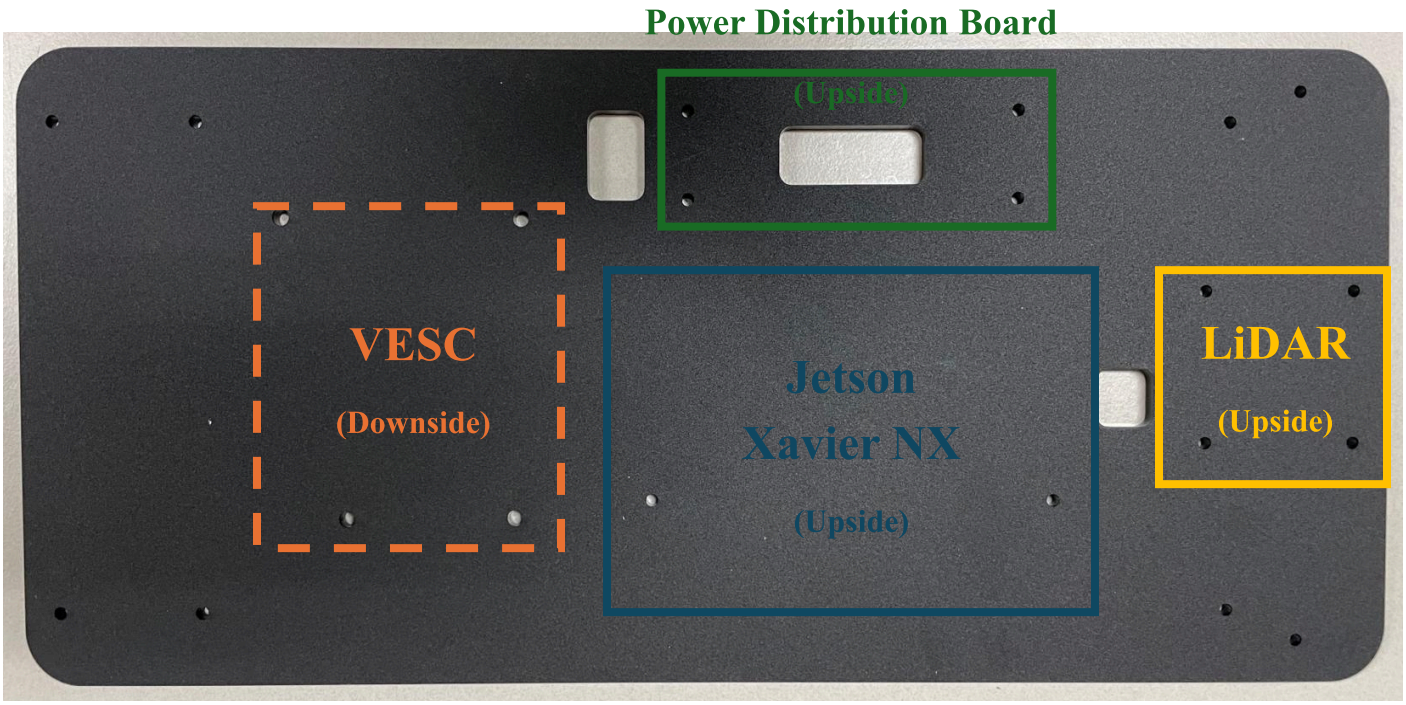
3-1. ESC

3-2. Power distribution board

3-3. Jetson Xavier NX board

3-4. Hokuyo Lidar

3-0. Upper deck configuration

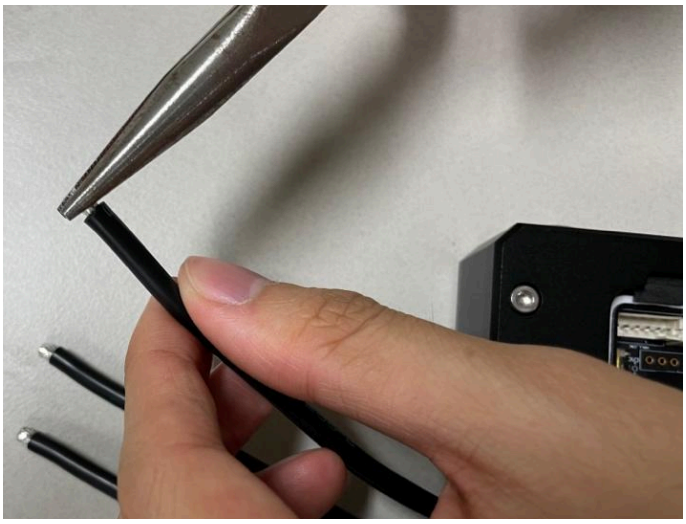


3-0. Attach the chassis connecting components. (M3 16mm screw x 8)

3-1. [ESC assembly process]

3-1-1. Attach gold connectors to ESC(female).

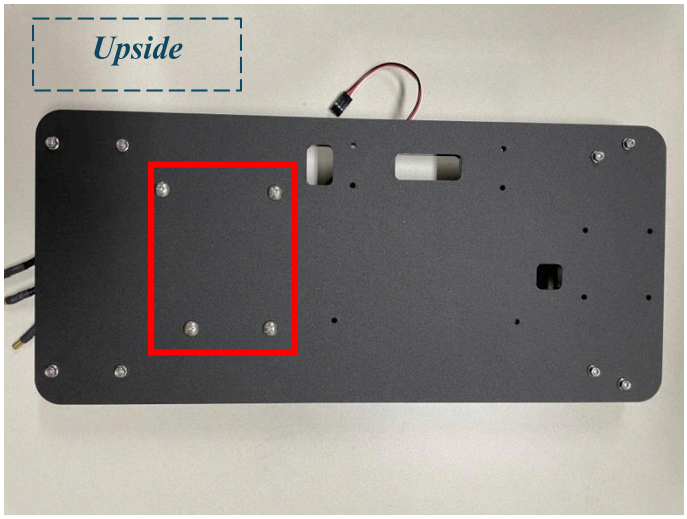
3-1-2. Use pliers to crimp the end of the ESC and fit it snugly into the gold connector.

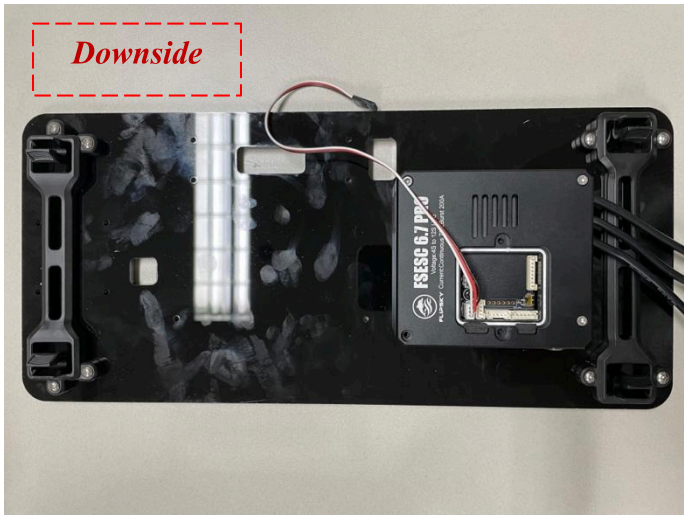


3-1-3. Connect them with 4mm heat shrink tube.

3-1-4. Attach the ESC to the upper deck (M4 10mm x 4) and connect the PWM cable to the

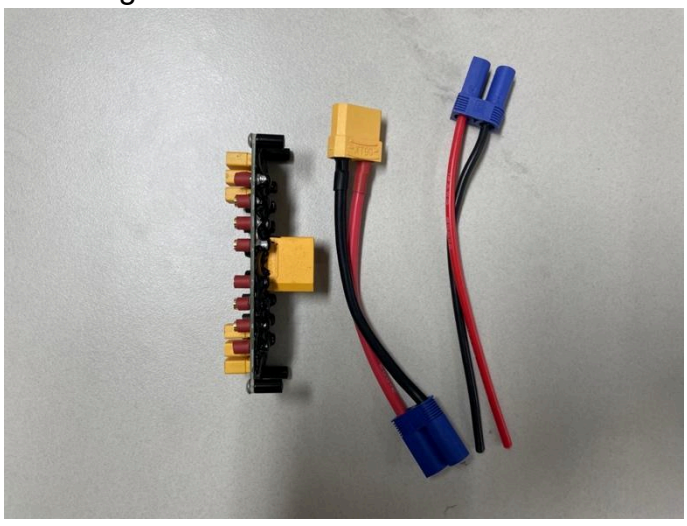
ESC.





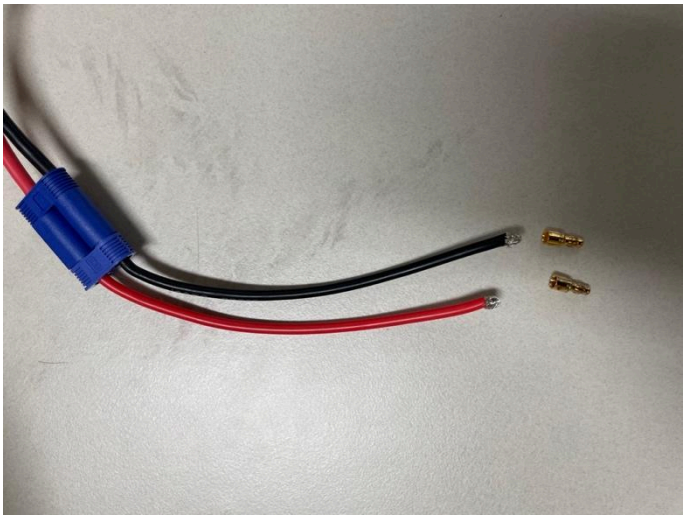
3-2. [Power distribution board assembly process]

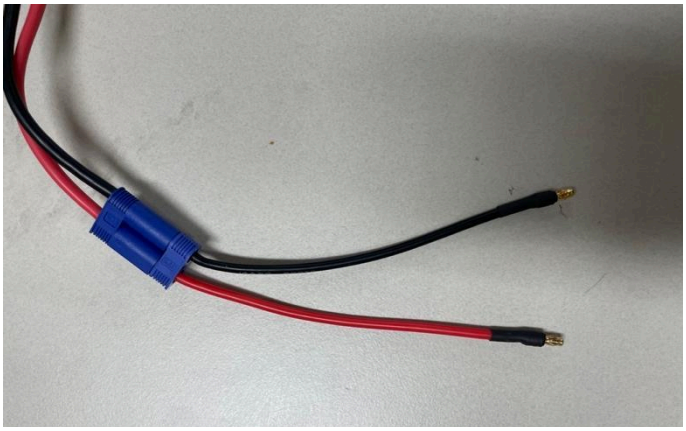
3-2-0. Ingredients



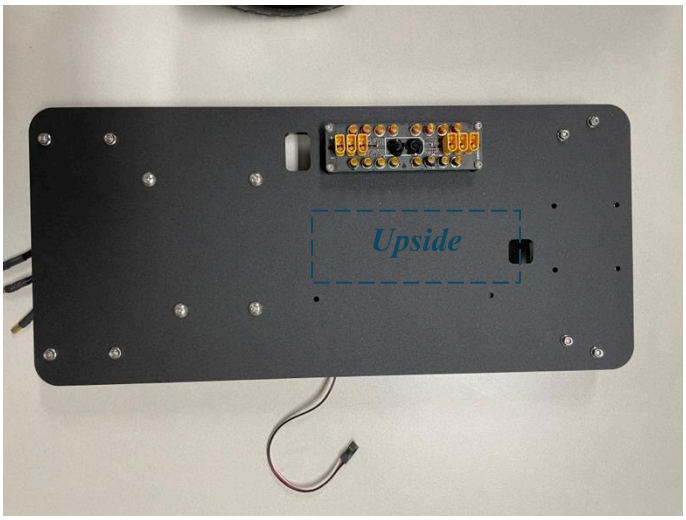
3-2-1. Connect EC5 longing cable to EC5 to XT90 cable.

3-2-2. After coupling the gold connector(male), use a 4mm heat shrink tube to seal the connection.

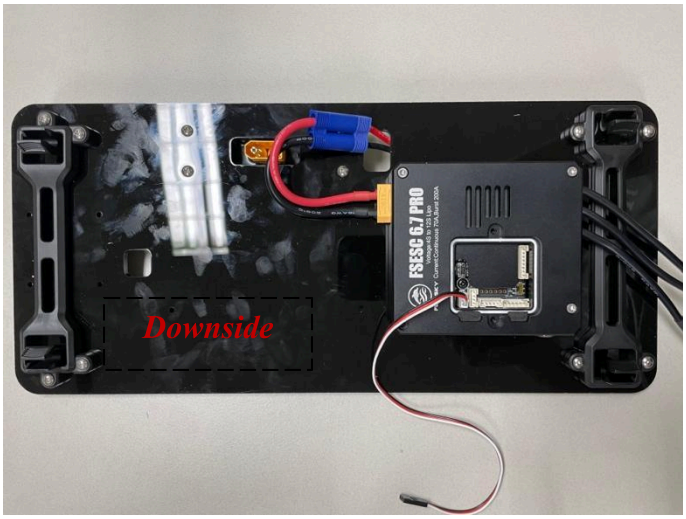




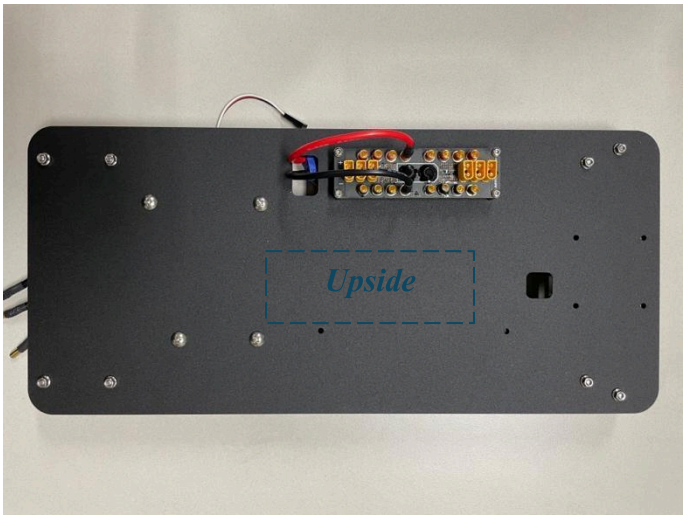
3-2-3. Attach power distribution board to the upside of the upper deck.



3-2-4. Connect the XT90 connector to the ESC.

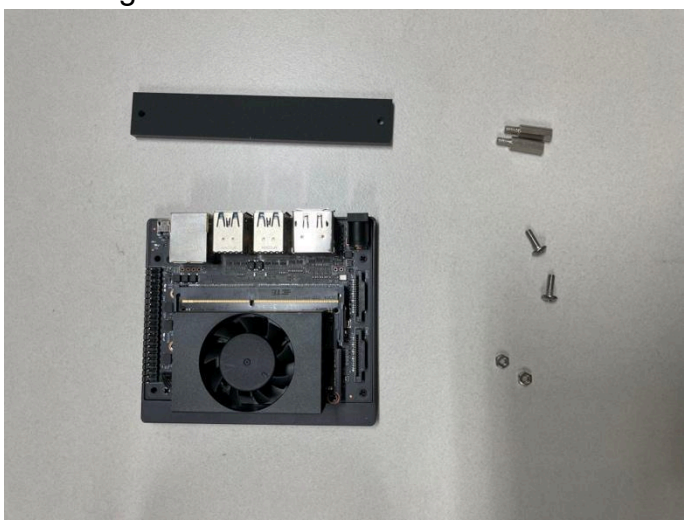


3-2-4. Connect the gold connector to the power distribution board.



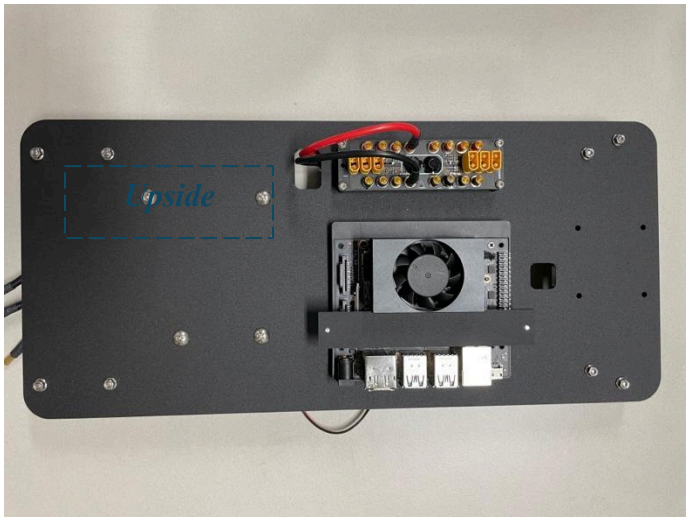
3-3. [Jetson board assembly process]

3-3-0. Ingredients



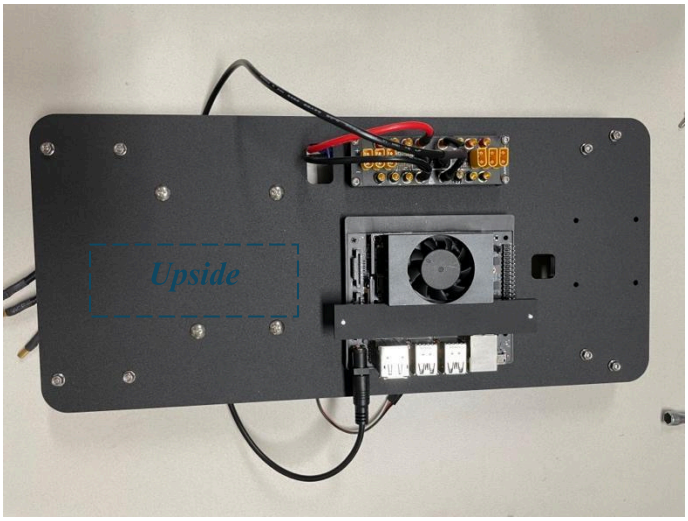
3-3-1.Assemble the ‘bracket’: fasten an M3 10 mm screw to the 15 mm spacer.

3-3-2. Fix Jetson Xavier NX board to the upper deck with bracket. (M3 10mm screw x2 + bracket)

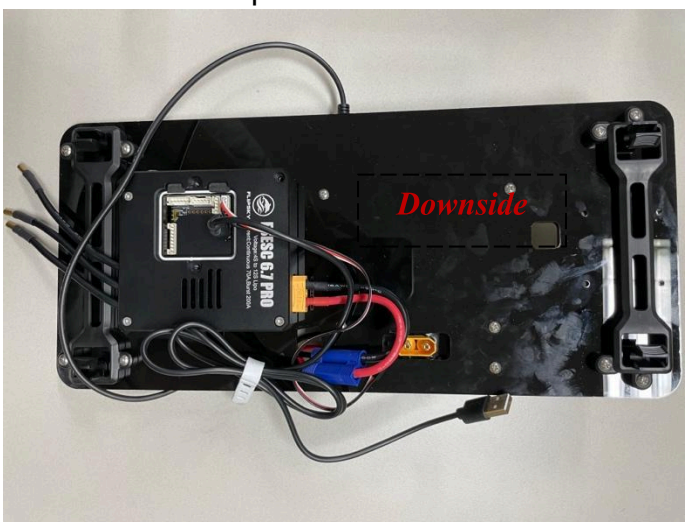


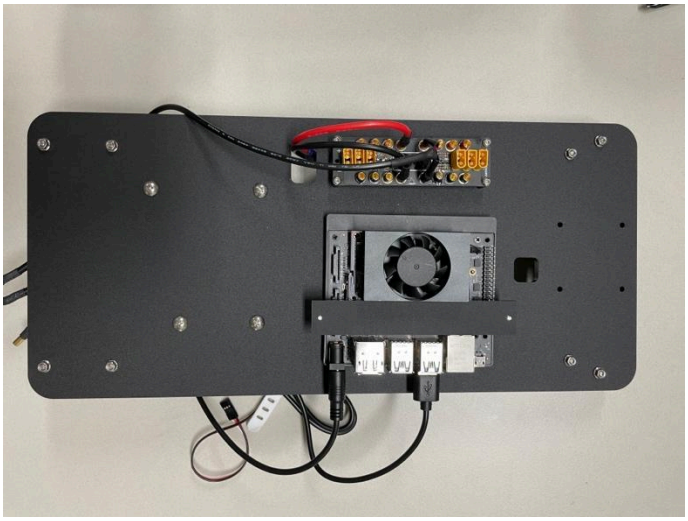
3-3-3. Pass the Jetson power cable through the stripped gold connector (male), bundle them together, and secure with 4mm heat shrink tubing and insulation tape.

3-3-4. Connect the power board to the Jetson board with above power cable.



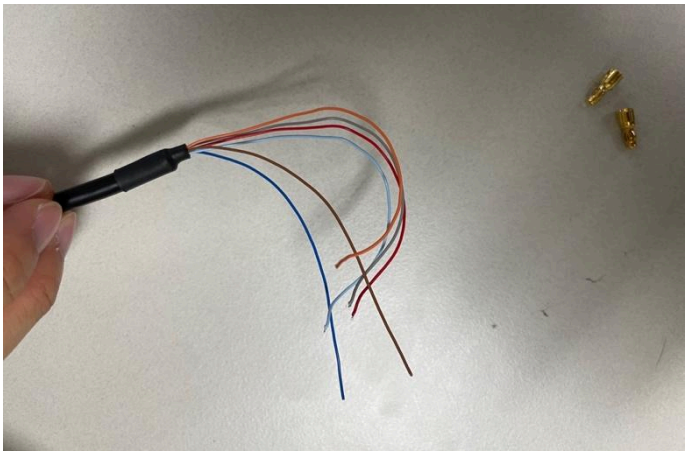
3-3-5. Connect the ESC and Jetson board using a USB to micro 5-pin cable.



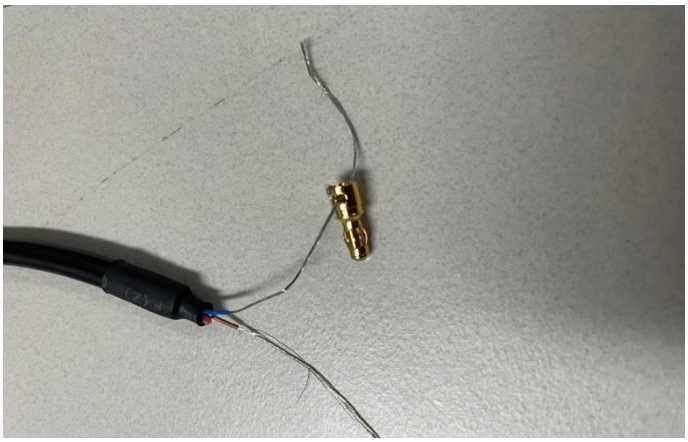


3-4. [Lidar assembly process]

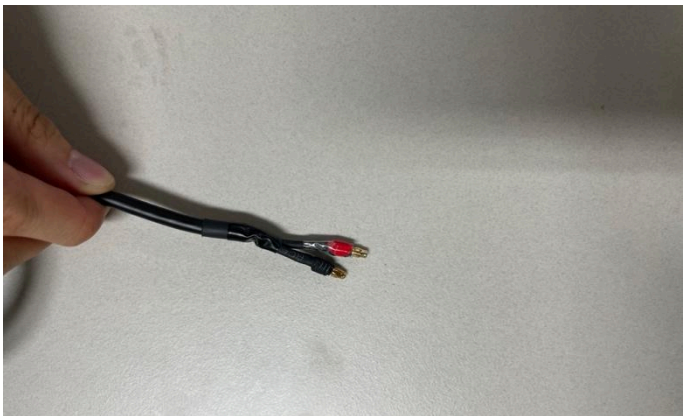
3-4-1. Cut and remove the insulation from all cables except the blue and brown ones in the Lidar power section. Blue (-) and brown (+) are the correct wires to remain.



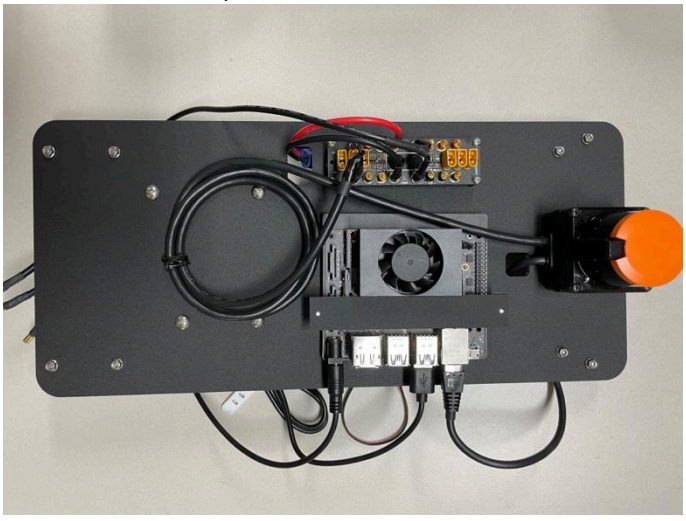
3-4-2. Insert 6mm and then 4mm heat shrink tubing in sequence (not shown in the photo, but necessary), pass through the brass connector and hole, and tie multiple times.



3-4-3. Secure with heat shrink tubing and insulation tape.

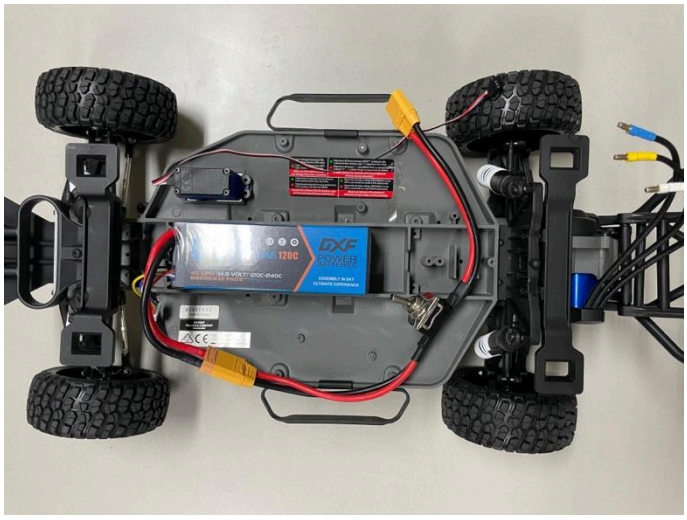


3-4-4. Attach the Lidar to the upper deck. (M3 10mm screw x2)

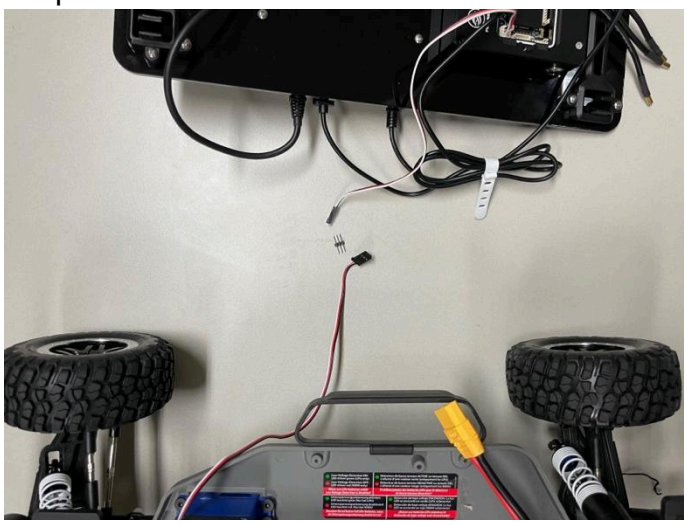


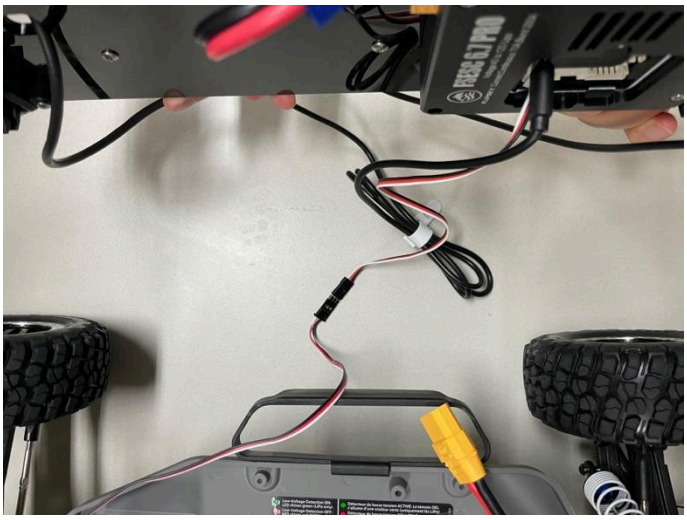
4. Upper deck and chassis assembly

4-1. Connect the LiPo battery and switch cable with XT90 connector.

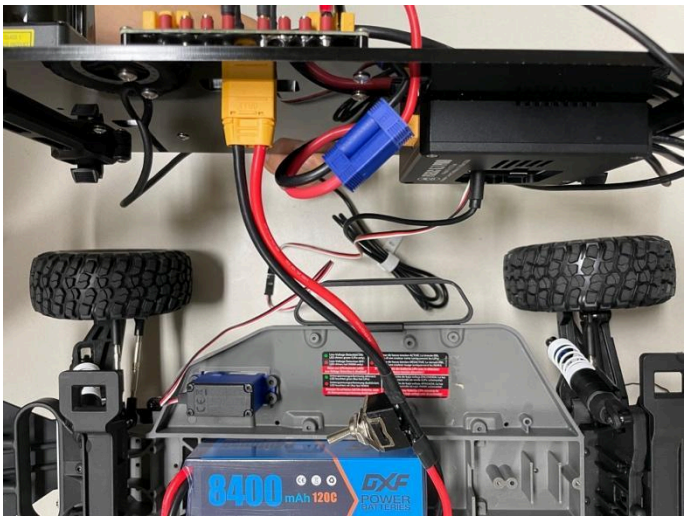


4-2. Connect the PWM cable, which is connected to the ESC, to the servo motor using a 3-pin connector.

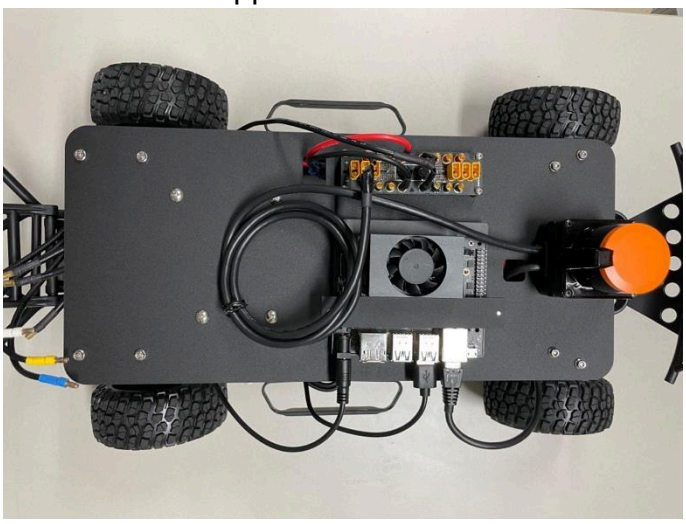




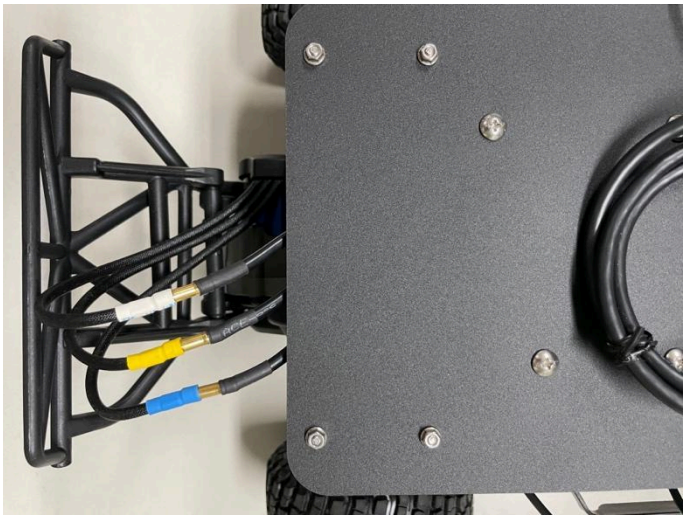
4-3. Connect the XT90 connector from the power board on the upper deck to the XT90 switch cable.



4-4. Attach the upper deck to the chassis.

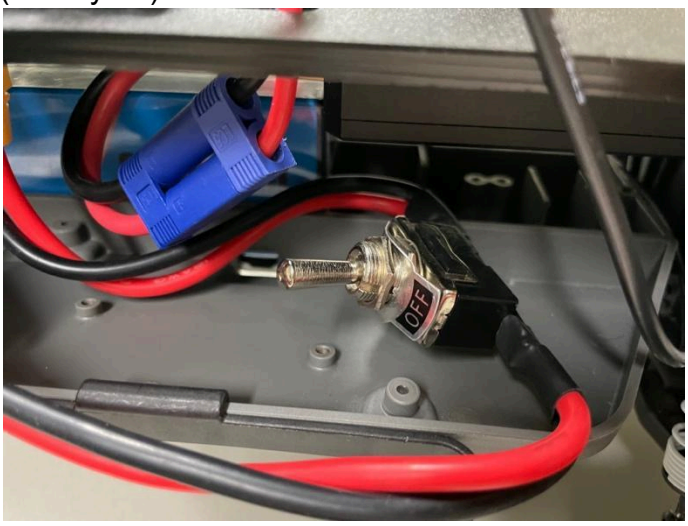


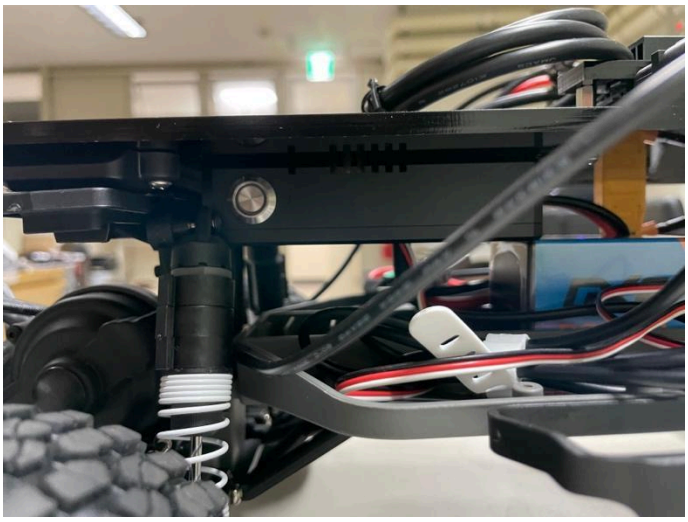
4-5. Connect the three-phase motor wires to the ESC in any order. The wire colors do not matter.



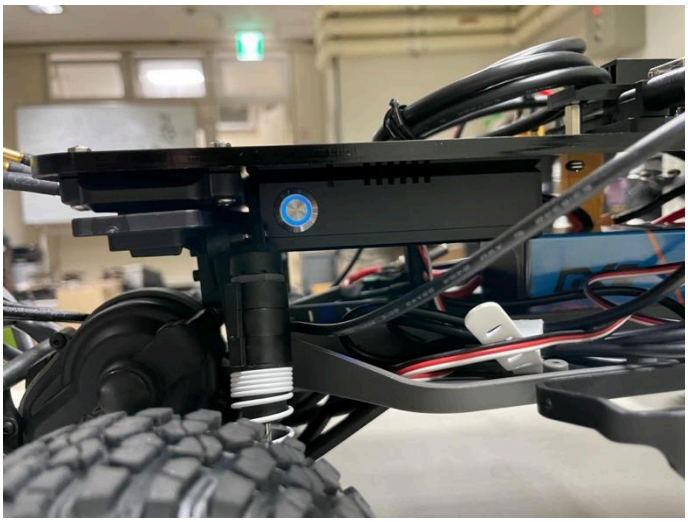
4-6. When you turn on the ESC power switch, the power is supplied to the Lidar, Jetson board, and ESC. The Lidar emits a beep- sound, the Jetson board's LED turns green, and the ESC's power LED turns blue, indicating that power is successfully supplied to all components.

(initially off)





(turn on)



Completed!

Please proceed to the VESC setup and Jetson Xavier NX board setup documents.

5. Supplementary

5-1. [Copyright]

This manual was written by Junseok Kim, Geunje Cheon, and Hyeokjin Kwon from the Robot Learning Laboratory (RLLAB) at Seoul National University. The copyright of this document belongs to RLLAB.
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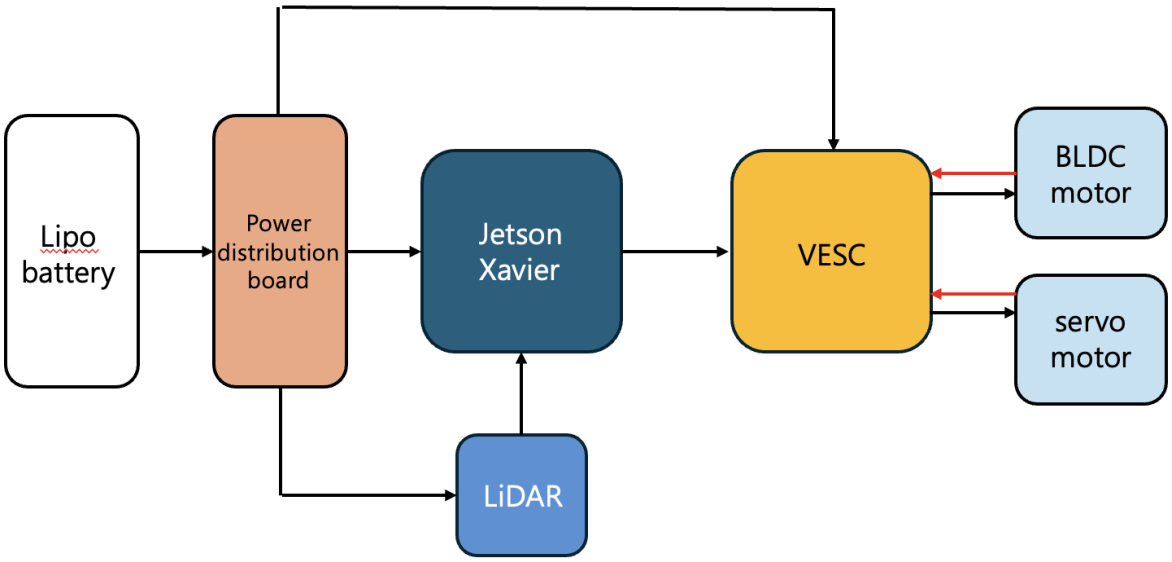
geunje.cheon@rllab.snu.ac.kr
hyeokjin.kwon@rllab.snu.ac.kr

5-1. [Schematic of our RC car]



The 4S LiPo battery connected to the power distribution board powers the LiDAR, Jetson Xavier NX board, and VESC (ESC). The Jetson Xavier NX board uses inputs from the LiDAR and odometry calculations from the ROS VESC package to output the necessary speed and direction for autonomous driving. The VESC adjusts the BLDC sensorless motor and servo motor according to the desired speed and direction received from the Jetson board.

For this setup, it is necessary to install the ROS program on the Jetson board and use the VESC package. The VESC setup tool should be used to adjust PID coefficients and various settings in advance. Detailed instructions for these processes can be found in the VESC setup manual and Jetson Xavier NX board setup manual. Please refer to these documents for further guidance.